

ITHENA Technical Evaluation Assignment

PROJECT 3: Snowflake- Cost & Query Performance

QUESTION 16:

WorkspacesDatabases+Q

My Workspace

+ Add new

QUESTION 16.sql

HomeQUESTION 16.sql

ACCOUNTADMIN • COMPUTE_WH (X-Small)Choose databaseShare

```
1 SELECT
2 LEFT(query_text, 50) AS query_pattern,
3 COUNT(*) AS times_run,
4 SUM((execution_time / 1000.0 / 3600.0) *
5 CASE warehouse_size
6 WHEN 'Small' THEN 2
7 WHEN 'Medium' THEN 4
8 WHEN 'Large' THEN 8
9 ELSE 1
10 END ) AS estimated_credits
11 FROM SNOWFLAKE.ACCOUNT_USAGE.QUERY_HISTORY
12 WHERE start_time >= DATEADD('day', -30, CURRENT_TIMESTAMP())
13 AND execution_status = 'SUCCESS'
14 GROUP BY LEFT(query_text, 50)
15 ORDER BY estimated_credits DESC
16 LIMIT 10;
```

Results (3 minutes ago)

TableChartPivot

10 rows2.1s

	QUERY_PATTERN	# TIMES_RUN	# ESTIMATED_CREDITS
	ALTER WORKSPACE "USERS"."PUBLIC"."DEF... +8 more	10.0% 10.0%	10.0%
1		1	0.000024166667
2	SELECT LEFT(query_text, 50) AS query_pattern,	21	0.000934444444
3	CREATE WORKSPACE IF NOT EXISTS "USERS"."PUBLIC";	10	0.000934444444
4	SELECT warehouse_name, COUNT(*) AS thrashing_hour	1	0.000209722222
5	PUT 'file://Untitled.sql' 'snow://workspace/USERS.	1	0.000181111111
6	PUT 'file://viewedFiles.json' 'snow://workspace/US	1	0.000128611111
7	SELECT SYSTEM\$GET_WINTERFEST_CONFIGS()	21	0.000071111110
8	PUT 'file://viewedFiles.json' 'snow://workspace/US	13	0.000064722221

RESULTS:

Results (4 minutes ago)

TableChartPivot

10 rows2.1s

	QUERY_PATTERN	# TIMES_RUN	# ESTIMATED_CREDITS
	ALTER WORKSPACE "USERS"."PUBLIC"."DEF... +8 more	10.0% 10.0%	10.0%
1		1	0.000024166667
2	SELECT LEFT(query_text, 50) AS query_pattern,	21	0.000934444444
3	CREATE WORKSPACE IF NOT EXISTS "USERS"."PUBLIC";	10	0.000934444444
4	SELECT warehouse_name, COUNT(*) AS thrashing_hour	1	0.000209722222
5	PUT 'file://Untitled.sql' 'snow://workspace/USERS.	1	0.000181111111
6	PUT 'file://viewedFiles.json' 'snow://workspace/US	1	0.000128611111
7	SELECT SYSTEM\$GET_WINTERFEST_CONFIGS()	21	0.000071111110
8	PUT 'file://viewedFiles.json' 'snow://workspace/US	13	0.000064722221
9	SELECT SYSTEM\$GET_WINTERFEST_CONFIGS()	1	0.000062500000

QUESTION 17:

The screenshot shows a SQL IDE interface. On the left is a sidebar with navigation icons. The main area is split into a query editor and a results pane. The query editor contains the following SQL code:

```
1 SELECT
2   warehouse_name,
3   COUNT(*) AS thrashing_hours,
4   SUM(start_count) AS total_starts_in_thrashing_hours,
5   AVG(avg_run_seconds) AS avg_run_length_seconds
6 FROM (
7   SELECT
8     warehouse_name,
9     DATE_TRUNC('hour', start_time) AS hour_window,
10    COUNT(*) AS start_count,
11    AVG(DATEDIFF('second', start_time, end_time)) AS avg_run_seconds
12 FROM SNOWFLAKE.ACCOUNT_USAGE.WAREHOUSE_METERING_HISTORY
13 WHERE start_time >= DATEADD('day', -30, CURRENT_TIMESTAMP())
14 GROUP BY warehouse_name, DATE_TRUNC('hour', start_time)
15 ) hourly_activity
16 WHERE start_count >= 3
17 AND avg_run_seconds < 300
18 GROUP BY warehouse_name
19 ORDER BY thrashing_hours DESC;
```

The results pane, titled "Results (2 minutes ago)", shows a table with the following columns: WAREHOUSE_NAME, THRASHING_HOURS, TOTAL_STARTS_IN_THRASHING_HOURS, and AVG_RUN_LENGTH_SECONDS. The table is empty, and a message states "Query produced no results".

RESULTS:

This is a close-up view of the results pane from the previous screenshot. It shows the table structure and the empty result set.

WAREHOUSE_NAME	THRASHING_HOURS	TOTAL_STARTS_IN_THRASHING_HOURS	AVG_RUN_LENGTH_SECONDS
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Query produced no results